

Linyi (Tiger) Hou

✉ linyh2@illinois.edu | github.com/TigerHou2 | Permanent Resident

EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign Jun 2021 - May 2025
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign Aug 2017 - May 2021

SKILLS

Technical celestial navigation, astrodynamics, Kalman filtering, modeling & simulation, data analysis
Collaboration technical/proposal writing, MS Office suite, L^AT_EX, Git, GitHub, Trello
Programming proficient: Python, C/C++, MATLAB | beginner: Simulink, Javascript, ADQL

EXPERIENCE

Graduate Researcher, University of Illinois Urbana-Champaign Jun 2021 - present

- ❖ Developed several novel spacecraft navigation and attitude determination techniques.
- ❖ Assessed navigation performance through simulations and Monte Carlo analysis.
- ❖ Presented research results at conferences and in peer-reviewed journal publications.

Teaching Assistant, University of Illinois Urbana-Champaign Jun 2021 - May 2024

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Software Engineering Intern, Collins Aerospace May 2019 - Aug 2019

- ❖ Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- ❖ Automated weekly builds and integration in TortoiseSVN using Python.

Team Lead/Advisor, NASA's RASC-AL Competition 2019, 2022 - 2023

- ❖ Advised two student teams who developed requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.
- ❖ Led a team of 10+ students to develop a reusable crewed lunar lander concept.

HONORS AND AWARDS

Yee Fellowship 2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship 2024
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference 2024
American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship 2020
American Institute of Aeronautics and Astronautics (AIAA)